Out of Work
Forecasting and Explaining U.S. Unemployment Through Macroeconomic Indicators

M4 Write-Up Rough Draft

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DATA 510 Capstone - Summer 2026

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August 2026

Project scope: U.S. monthly data, April 1956-December 2025

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GitHub Repository

- https://github.com/manishkallu01-wq/DATA510-Session-2/tree/main

GitHub Project Board

- https://github.com/users/manishkallu01-wq/projects/1

Discord Category

- https://discord.com/channels/1277725100816203942/1508588739063054376

Studio Session: 2

Studio Formation Date: May 25, 2026

Abstract

This project evaluates whether publicly available macroeconomic indicators can forecast the U.S. unemployment rate three, six, and twelve months in advance and which original macroeconomic indicators show the strongest historical relationship with unemployment. Six forecasting models were estimated on monthly data covering April 1956 through December 2025 using 35 engineered predictors. Models were selected with validation RMSE and evaluated on a post-1999 chronological test period. Ridge Regression was selected at three months, while Extra Trees was selected at six and twelve months. Test performance declined from RMSE 1.020 and R^2 0.720 at three months to RMSE 1.729 and R^2 0.205 at twelve months. Research Question 2 was evaluated separately through Pearson and Spearman lead-lag correlations between unemployment and five external indicators: inflation, the federal funds rate, GDP growth, consumer sentiment, and recession status. Consumer sentiment had the strongest overall relationship, with Pearson r strengthening from -0.351 in the same month to -0.496 at twelve months; recession status ranked second. GDP growth showed a modest negative relationship, while inflation and the federal funds rate became more positive at longer leads. Historical-period results show that these relationships are not constant across every regime, especially after 2020. The forecasting results support short-horizon use as an early-warning aid, while the correlation results describe historical association rather than causation.

> **Main finding**
> Public macroeconomic data support useful three-month forecasts and moderately informative six-month forecasts. In the separate historical-correlation analysis, consumer sentiment is the strongest overall external relationship with future unemployment, although the relationship is not stable in every period.

Report map

- Sections 1 – 3 define the problem, stakeholders, and related forecasting literature.
- Sections 4 – 6 document the data pipeline, responsible-use considerations, and time-based evaluation design.
- Section 7 reports the model results and presents the selected-model forecasting and correlation figures.
- Sections 8 – 11 answer both research questions, discuss limitations, provide recommendations, and document reproducibility.

1. Introduction: Problem and Research Questions

The unemployment rate is one of the most visible measures of labor-market health, but it is also a lagging and shock-sensitive outcome. Policymakers, employers, workforce agencies, and investors benefit when deterioration can be identified before it becomes fully visible in headline labor statistics. The practical challenge is not merely to fit historical unemployment; it is to determine how far ahead a model can forecast without relying on future information and whether commonly available indicators add value beyond unemployment history itself.

This project builds a reproducible forecasting workflow around six public macroeconomic series and a set of engineered lag, growth, momentum, and stress features. The analysis treats the three-, six-, and twelve-month horizons as distinct forecasting problems. That distinction matters because a model that is useful one quarter ahead may be too smooth or unstable one year ahead. The study also separates model selection from final testing, which prevents the choice of a model from being driven by the same period used to report performance.

1.1 Research questions

1. Can future U.S. unemployment be forecast three, six, and twelve months ahead using publicly available macroeconomic indicators?
2. Which macroeconomic indicators show the strongest and most consistent historical relationship with unemployment across economic environments and business cycles?

1.2 Analytical contribution

The contribution is a horizon-specific comparison rather than a single overall accuracy claim. All six algorithms are evaluated under the same chronological test design, against the same persistence and unemployment-only baselines, and with the same error metrics. The report therefore distinguishes three questions that are often conflated: which model was selected without using the test set, which model happened to obtain the lowest test error, and whether the selected full model improves on a simpler benchmark. Research Question 2 is evaluated separately through contemporaneous and lead-lag correlations using the five external indicators, so machine-learning feature importance is not mistaken for historical correlation.

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2. Impact and Stakeholders

A credible short-horizon unemployment forecast can support planning before labor-market deterioration is fully reflected in official releases. The model is not intended to replace economic judgment. Its value is to organize multiple indicators into a consistent historical signal and document how forecast error increases with the forecast horizon.

Table 1. Stakeholders, uses, and responsible interpretation

Stakeholder	Potential use	Required caution
Federal, state, and local policymakers	Scenario planning for benefits, fiscal response, and labor programs	Forecasts should be combined with real-time administrative and survey data.
Workforce agencies and universities	Anticipate demand for retraining, placement, and student support	National averages may not represent local labor markets or industries.
Businesses and finance teams	Workforce, demand, and risk planning	The model does not identify firm-level exposure or causal effects.
Economists and researchers	Benchmark public-data forecasting methods and examine indicator relevance	Retrospective recession features limit real-time interpretation.
Workers and the public	Understand broad labor-market direction	Outputs must not be presented as certain predictions of individual job loss.

3. Related Work and Background

Macroeconomic forecasting often combines lagged outcomes with indicators of production, prices, interest rates, and expectations. Broad predictor sets can improve forecasting performance, while standardized monthly databases such as FRED-MD support consistent macroeconomic analysis (*Stock & Watson, 2002; McCracken & Ng, 2016*). Consumer confidence may also contain information about future economic activity (*Barsky & Sims, 2012*). This project therefore uses a compact indicator set to forecast unemployment and examine lead–lag relationships with inflation, interest rates, GDP growth, consumer sentiment, and recession status.

NBER recession dates are used only for historical evaluation because turning points are identified retrospectively (*National Bureau of Economic Research, n.d.).* The model comparison includes linear, regularized, bagging, and boosting approaches to test whether nonlinear flexibility improves forecasting relative to simpler, more interpretable benchmarks.

Why a smaller indicator set?
The objective is not to reproduce a very large macroeconomic forecasting database. The six-source design is easier to audit, explain, refresh, and connect to policy narratives while still spanning labor conditions, prices, monetary policy, output, expectations, and recession regimes.

4. Data and Data Engineering

4.1 Source data

All source series were obtained through the Federal Reserve Economic Data system or its contributing institutions. Monthly observations were aligned to a common calendar. Gross domestic product, originally quarterly, was converted to a monthly representation before merging. The final modeling period begins in April 1956, when the required indicators and engineered histories are jointly available, and ends in December 2025.

Table 2. Public macroeconomic data sources

Series	Concept	Source organization	Native frequency	Role
UNRATE	Unemployment rate	U.S. Bureau of Labor Statistics	Monthly	Target, persistence benchmark, labor lags
CPIAUCSL	Consumer price index	U.S. Bureau of Labor Statistics	Monthly	Inflation and price-pressure features
FEDFUNDS	Effective federal funds rate	Federal Reserve Board	Monthly	Monetary-policy and real-rate features
GDP	Gross domestic product, current dollars	U.S. Bureau of Economic Analysis	Quarterly	Growth, lag and momentum features
UMCSENT	Consumer sentiment	University of Michigan	Monthly	Expectations and confidence deterioration
USREC	NBER recession indicator	National Bureau of Economic Research	Monthly	Historical regime indicator and segmentation

4.2 Pipeline and quality controls

1. Downloaded and standardized all source series using a consistent date column and numeric data types.
2. Aligned quarterly GDP to monthly frequency by carrying each quarterly value across the three months of its quarter.
3. Merged all six macroeconomic indicators using the monthly date key.
4. Created lagged, rate-of-change, momentum, ratio, stress, and future-unemployment target variables.
5. Sorted the data chronologically and verified valid formats, no duplicate dates, no infinite values, and no unexpected missing predictor values.
6. Exported the validated processed dataset as ‘capstone_capstone_plus_final.xlsx’. The final dataset contains **837 rows and 40 columns** and serves as the common analysis-ready input for both research questions.

4.3 Final analytical dataset

Table 3. Dataset profile and validation checks

Audit item	Result
Rows	837
Columns	40
Numeric predictors used	35
Date range	1956-04-01 to 2025-12-01
Duplicate dates after cleaning	0
Infinite numeric values after cleaning	0
Intentional missing targets	3 at 3M, 6 at 6M, and 12 at 12M

Missing target values occur only at the end of the series because future unemployment is not yet observable for those forecast horizons.

4.4 Feature engineering

The 35 predictors include the contemporaneous unemployment rate; unemployment lags at one, three, six, and twelve months; unemployment changes and momentum; inflation rates and lags; GDP growth and momentum; federal-funds-rate lags; real interest rate; consumer-sentiment lags and deterioration measures; and composite ratios or stress indexes. Separate targets shift the unemployment rate forward by three, six, or twelve months.

Table 4. Predictor groups used in the submitted modeling run

Feature group	Examples	Purpose
Labor persistence	UnemploymentRate, Lag1, Lag3, Lag6, Lag12	Represent inertia and delayed adjustment in labor markets.
Labor acceleration	Change1M, Change3M, UnemploymentMomentum	Capture emerging deterioration or improvement.
Prices and policy	InflationRateYoY, inflation lags, FEDFUNDS lags, RealInterestRate	Represent price pressure and monetary restraint.
Output	GrossDomesticProduct, GDPGrowthYoY, growth lags and momentum	Measure expansion, contraction, and production trends.
Expectations	ConsumerSentiment, sentiment lags, confidence deterioration	Represent household expectations before realized activity changes.
Regime/composites	RecessionIndicator, EconomicStressIndex, RecessionRiskScore	Describe historical economic regimes and combined stress.

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5. Ethics and Responsible Use

The project uses aggregate public economic data and contains no personal records, protected attributes, or individual employment histories. Privacy risk is therefore low. The principal ethical risks arise from interpretation: national forecasts may be mistaken for individual predictions, a model may be presented as more certain than its performance supports, or retrospective information may be described as if it were available in real time.

Three safeguards are central. First, performance is reported on an untouched chronological test period and compared with simple baselines. Second, the document reports weak twelve-month performance rather than selecting only favorable results. Third, it treats the recession indicator and recession-risk features as historical descriptors. NBER turning points are dated retrospectively, so any model containing those features should be described as a historical backtest rather than a deployable real-time early-warning system.

The EconomicStressIndex also includes unemployment-related information. Its feature importance must not be interpreted as evidence that an independent external variable predicts unemployment. Likewise, permutation importance identifies predictive dependence within a fitted model, while the separate correlation analysis identifies historical association. Neither method establishes that changing interest rates, sentiment, inflation, GDP, or recession status would causally change unemployment by a corresponding amount.

Responsible-use statement
The forecasts are appropriate for research, scenario planning, and comparative model evaluation. They are not suitable as the sole basis for benefit allocation, hiring or layoff decisions, recession declarations, or claims about any individual worker.

6. Methods and Evaluation

6.1 Chronological design

Random train-test splitting would mix observations from different historical periods and allow later economic regimes to influence earlier forecasts. The project therefore uses date-based development and test periods. Validation begins in January 1989. Testing begins in January 1999 and continues through the latest target available for each horizon. An embargo equal to the forecast horizon separates the development target dates from the next split, preventing a training label from extending into validation or test time.

Table 5. Horizon-specific chronological split

Horizon	Initial train rows	Validation rows	Final pre-test fit rows	Test rows	Test origin range
3M	390	117	510	321	Jan 1999 — Sep 2025
6M	387	114	507	318	Jan 1999 — Jun 2025
12M	381	108	501	312	Jan 1999 — Dec 2024

6.2 Model set and selection rule

The six-model comparison includes two linear estimators and four tree ensembles. Linear Regression serves as an unregularized benchmark. Ridge Regression regularizes correlated predictors (*Hoerl & Kennard, 1970*). Random Forest and Extra Trees average randomized decision trees (*Breiman, 2001; Geurts et al., 2006*), while Gradient Boosting and XGBoost sequentially reduce prediction errors (*Friedman, 2001; Chen & Guestrin, 2016*).
For each horizon, all models are fitted on the initial training period and evaluated on validation data. The model with the lowest validation RMSE, with validation MAE as a tiebreaker, is selected. Models are then refitted on the full pre-test period and evaluated once on untouched test data. The selected model is retained even if another algorithm later achieves a lower test RMSE, preventing the test period from becoming a second validation set.

6.3 Evaluation measures and baselines

Mean absolute error (MAE) reports the average absolute miss in unemployment percentage points. Root mean squared error (RMSE) penalizes occasional large errors more heavily. R^2 reports the share of test-period variation explained relative to predicting the test mean. Recall is not a valid metric for this task because the target is continuous rather than a positive/negative class.

Two baselines test whether the full feature set adds value. The persistence baseline assumes future unemployment equals the current unemployment rate. The unemployment-only linear baseline uses labor-market features without the broader macroeconomic set. A useful full model should improve on these benchmarks, not merely outperform another complex algorithm.

6.4 Historical correlation and lead–lag analysis

Research Question 2 was evaluated separately from the machine-learning analysis. Inflation, the federal funds rate, GDP growth, consumer sentiment, and recession status were compared with unemployment in the same month and three, six, and twelve months later. Pearson correlation measured linear association, while Spearman correlation provided a robustness check; for binary recession status, Pearson correlation is equivalent to point-biserial correlation. Mean absolute Pearson correlation summarized overall relationship strength across horizons. Historical-period comparisons used the six-month lead as the intermediate early-warning horizon, with the shorter 2020–2025 period interpreted cautiously. These results describe historical association, timing, and regime stability—not causation.

7. Results and Visual Evidence

The results are organized by forecast horizon because forecast quality, selected algorithm, and relevant predictors change materially with lead time. Each table reports all six algorithms. The highlighted row is the model selected using validation RMSE; it is not necessarily the row with the lowest test RMSE. The main report presents the selected-model forecast plots, while the complete model comparisons are available on GitHub.

7.1 Three-month horizon: strong short-term performance

Table 6. Three-month model performance

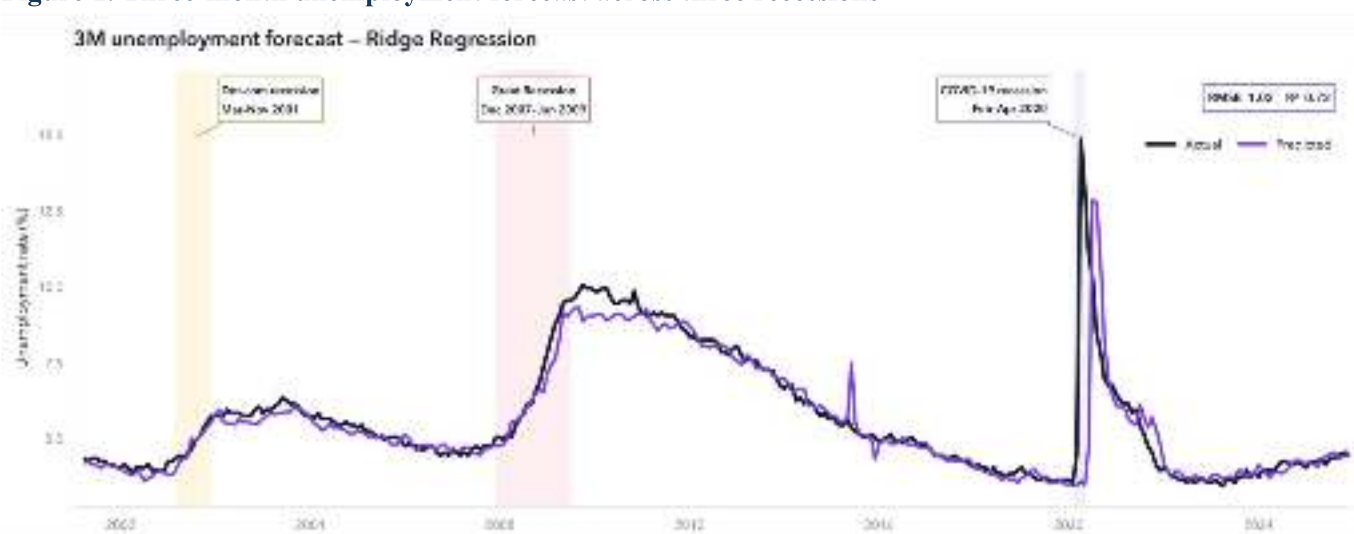
Model	Val. RMSE	Test MAE	Test RMSE	Test R^2	Selected
Ridge Regression	0.218	0.349	1.020	0.720	Yes
XGBoost	0.276	0.421	0.962	0.751	No
Gradient Boosting	0.269	0.410	0.964	0.750	No
Extra Trees	0.236	0.385	0.965	0.749	No
Linear Regression	0.334	0.386	1.059	0.698	No
Random Forest	0.317	0.646	1.114	0.666	No

The selected model is based on validation RMSE, not the lowest observed test RMSE.

Ridge Regression was selected because it obtained the lowest validation RMSE of 0.218; its validation R^2 was 0.946. On the test period, it achieved MAE 0.349, RMSE 1.020, and R^2 0.720. XGBoost and the boosting/tree ensembles produced slightly lower test RMSE values, but they were not selected because those test outcomes were unavailable at the decision point. Retaining Ridge is the correct out-of-sample procedure and favors a stable, interpretable linear model over post-hoc test optimization.

The selected model improved RMSE by 4.7% over persistence and 12.8% over the unemployment-only linear baseline. The plot shows close tracking during normal expansions and the gradual rise and recovery surrounding the Great Recession. The largest error occurs at the abrupt COVID-19 break: the model predicted 3.54% for April 2020 when actual unemployment reached 14.8%. It reacted only after the shock appeared in labor and macroeconomic inputs, demonstrating that the model is more effective for evolving cyclical deterioration than for sudden exogenous shutdowns.

Figure 1. Three-month unemployment forecast across three recessions



Selected Ridge Regression forecast. The shaded periods mark the 2001 recession, the Great Recession, and the COVID-19 recession.

Reported test performance: RMSE 1.02 and R^2 0.72.

7.2 Six-month horizon: moderate predictive value

Table 7. Six-month model performance

Model	Val. RMSE	Test MAE	Test RMSE	Test R²	Selected
Extra Trees	0.383	0.725	1.277	0.563	Yes
Ridge Regression	0.482	0.577	1.298	0.549	No
XGBoost	0.425	0.783	1.315	0.537	No
Gradient Boosting	0.457	0.795	1.330	0.526	No
Random Forest	0.476	0.856	1.388	0.484	No
Linear Regression	0.819	0.850	1.490	0.406	No

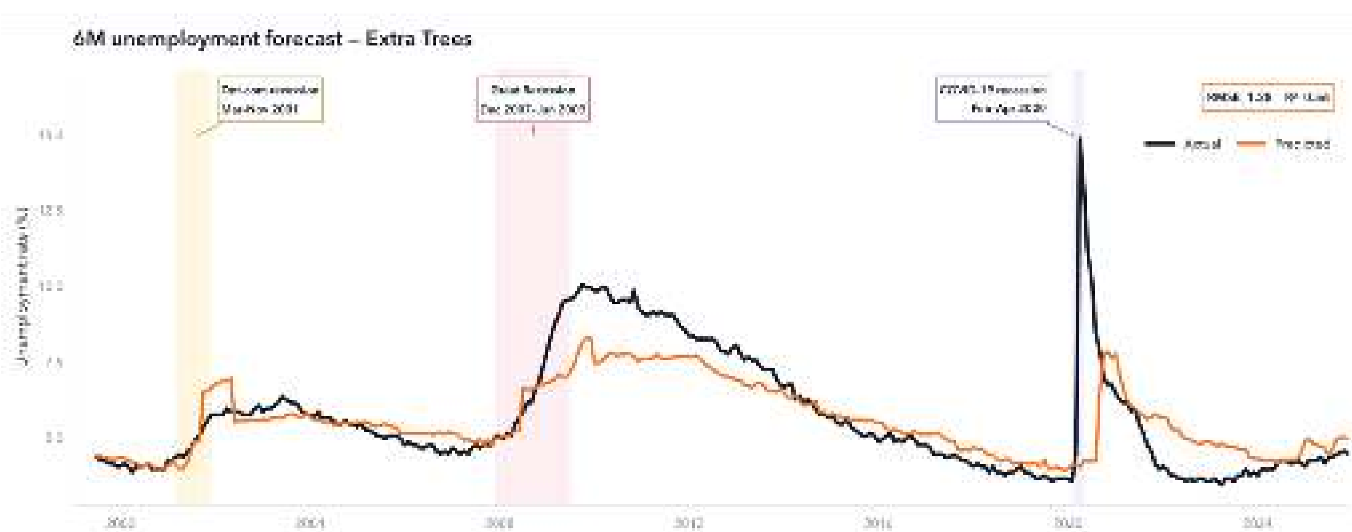
Extra Trees was selected before test evaluation because it had the lowest validation RMSE (0.383).

Extra Trees was selected for the six-month horizon with validation MAE 0.271, validation RMSE 0.383, and validation R^2 0.835. Test performance declined, as expected at a longer lead: MAE increased to 0.725, RMSE to 1.277, and R^2 fell to 0.563. The model reduced RMSE by 7.4% relative to persistence and 17.8% relative to the unemployment-only linear baseline, indicating that broader macroeconomic information helped reduce the most consequential errors.

The MAE comparison requires nuance. Persistence and the unemployment-only baseline obtained lower MAE than Extra Trees, even though their RMSE values were worse. This means the full model made somewhat larger typical errors in ordinary months but reduced some large misses enough to improve RMSE. Because early-warning applications are especially concerned with large labor-market deterioration, RMSE was retained as the selection metric. The report nevertheless presents both measures rather than claiming uniform dominance.

The six-month forecast captures the direction of the Great Recession but underestimates its peak and recovery path. It also fails to anticipate the initial COVID spike, remaining near 4% when actual unemployment moved into double digits. Compared with the three-month model, the prediction is smoother and slower to adapt, which is consistent with the weaker information available six months before a turning point.

Figure 2. Six-month unemployment forecast across three recessions



Selected Extra Trees forecast. The shaded periods mark the 2001 recession, the Great Recession, and the COVID-19 recession.

Test RMSE 1.28; R² 0.56.

7.3 Twelve-month horizon: limited incremental value

Table 8. Twelve-month model performance

Model	Val. RMSE	Test MAE	Test RMSE	Test R^2	Selected
Extra Trees	0.684	1.195	1.729	0.205	Yes
Ridge Regression	1.455	1.021	1.678	0.252	No
Random Forest	0.689	1.184	1.717	0.216	No
Gradient Boosting	0.774	1.167	1.750	0.186	No
XGBoost	0.797	1.233	1.831	0.108	No
Linear Regression	2.230	2.147	2.659	-0.879	No

Extra Trees was selected on validation data. Ridge had the lowest test RMSE but selecting it after test inspection would violate the evaluation design.

Extra Trees was selected using validation performance (RMSE 0.684; R^2 0.490), but its test performance was much weaker: MAE 1.195, RMSE 1.729, and R^2 0.205. The model improved RMSE by only 0.6% over persistence and was 0.9% worse than the unemployment-only linear baseline. The full macroeconomic feature set therefore did not provide a practically meaningful twelve-month advantage.

Ridge Regression happened to obtain the lowest twelve-month test RMSE (1.678), but its validation RMSE was 1.455 and validation R^2 was negative. That instability is precisely why test-period ranking cannot replace the pre-specified selection rule. Reporting Extra Trees as the selected model preserves methodological consistency, while the table transparently shows that the twelve-month result is sensitive to the chosen historical period.

The forecast plot is visibly smoother than actual unemployment. It underestimates the Great Recession peak, misses the COVID shock, and remains elevated relative to actual unemployment during portions of the post-2021 recovery. The twelve-month horizon is therefore best interpreted as a broad directional scenario rather than a precise early-warning forecast.

Figure 3. Twelve-month unemployment forecast across three recessions

Selected Extra Trees forecast. The shaded periods mark the 2001 recession, the Great Recession, and the COVID-19 recession.

Test RMSE 1.73; R^2 0.20.

7.4 Cross-horizon comparison and baseline evidence

Table 9. Selected models and RMSE improvement over baselines

Horizon	Selected model	MAE	RMSE	R^2	vs. persistence	vs. unemp.-only
3M	Ridge Regression	0.349	1.020	0.720	4.7%	12.8%
6M	Extra Trees	0.725	1.277	0.563	7.4%	17.8%
12M	Extra Trees	1.195	1.729	0.205	0.6%	-0.9%

Performance declines monotonically with the forecast horizon. R^2 falls from 0.720 at three months to 0.563 at six months and 0.205 at twelve months. The full model's baseline advantage is meaningful at three and six months but disappears at twelve months. This pattern directly answers the first research question: public macroeconomic indicators can support useful near-term unemployment forecasts, but the evidence does not support a strong claim of reliable twelve-month prediction.

7.5 Performance during highlighted recessions

Table 10. Selected-model error within the three highlighted recession periods

Horizon	Event	Months	Event MAE	Event RMSE
3M	2001 recession	9	0.16	0.21
3M	Great Recession	19	0.38	0.50
3M	COVID-19 recession	3	4.10	6.52
6M	2001 recession	9	0.43	0.57
6M	Great Recession	19	0.78	1.14
6M	COVID-19 recession	3	3.91	6.26
12M	2001 recession	9	0.53	0.63
12M	Great Recession	19	1.30	1.61
12M	COVID-19 recession	3	3.86	6.04

Errors were modest during the 2001 recession and increased during the Great Recession. COVID-19 produced a different scale of failure: event-specific RMSE exceeded 6 percentage points for every horizon because the initial labor-market collapse was not encoded in the preceding macroeconomic trajectory. The result is not evidence that the models are unusable; it defines the conditions under which they are most and least credible. They are better at forecasting persistent cyclical changes than discontinuities created by external public-health or policy shocks.

7.6 Historical indicator relationships and the second research question

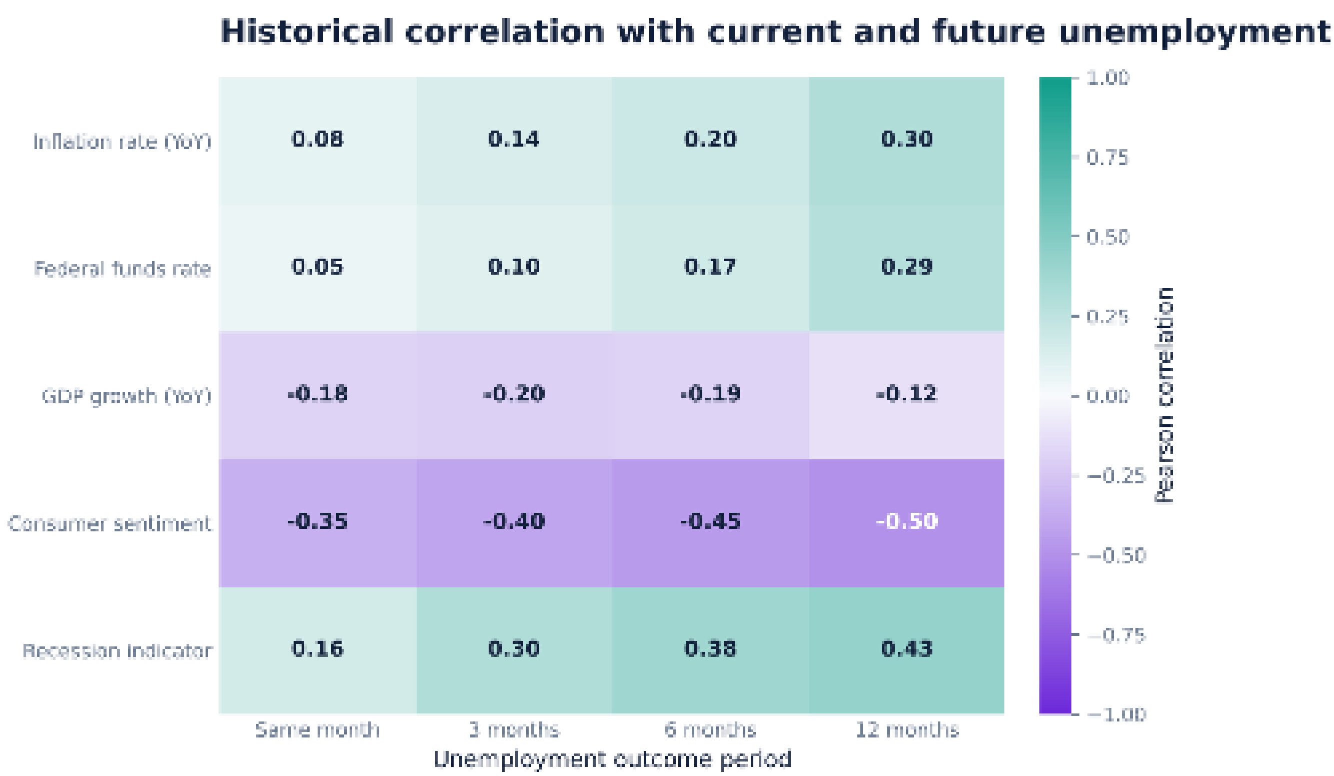
Table 11. Lead–lag correlations between macroeconomic indicators and current or future unemployment

Indicator	0M r	3M r	6M r	12M r	Mean |r|	Strongest lead
Consumer sentiment	-0.351	-0.399	-0.446	-0.496	0.423	12M (-0.496)
Recession indicator	0.159	0.298	0.378	0.429	0.316	12M (0.429)
Inflation rate (YoY)	0.076	0.137	0.195	0.304	0.178	12M (0.304)
GDP growth (YoY)	-0.184	-0.201	-0.186	-0.123	0.174	3M (-0.201)
Federal funds rate	0.052	0.104	0.166	0.287	0.152	12M (0.287)

Consumer sentiment showed the strongest full-sample relationship at every horizon, strengthening from r = −0.351 in the same month to r = −0.496 at twelve months. The negative direction indicates that weaker sentiment historically preceded higher unemployment. Spearman correlations followed a similar pattern, supporting the robustness of this result. Recession status ranked second, while inflation and the federal funds rate showed stronger positive relationships at longer leads. GDP growth had a modest negative relationship, strongest at three months.

Figure 4. Lead-lag correlations between external indicators and unemployment

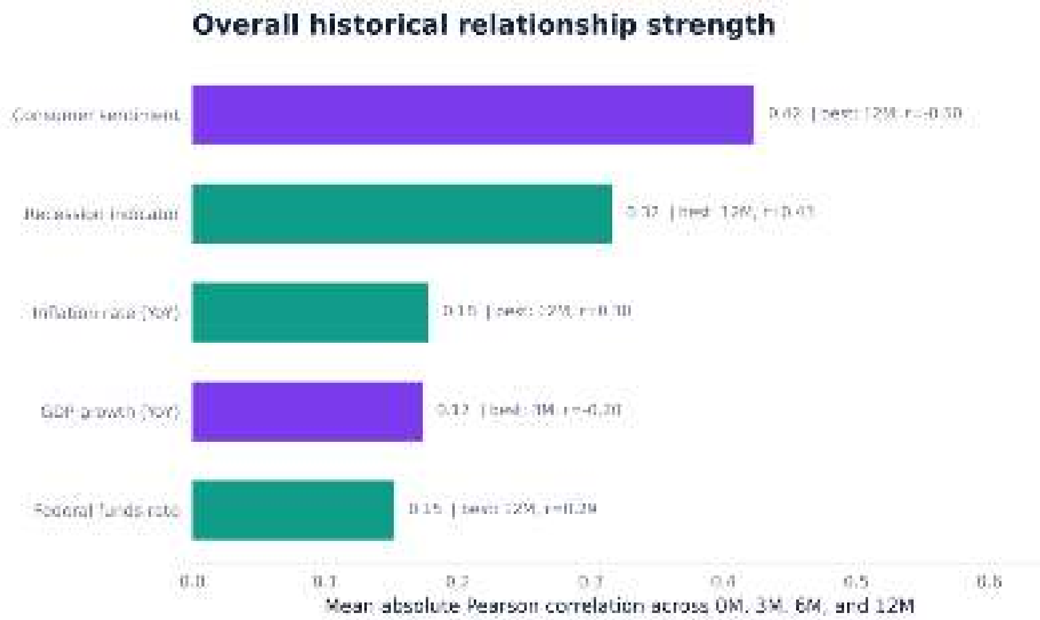
Consumer sentiment is the strongest full-sample relationship and becomes more negative as the unemployment horizon increases.



The heatmap directly answers the horizon component of Research Question 2. Positive values indicate that higher indicator values align with higher unemployment; negative values indicate an inverse relationship. The strongest cell is consumer sentiment at twelve months (r=-0.496).

Figure 5. Overall historical relationship strength across four lead periods

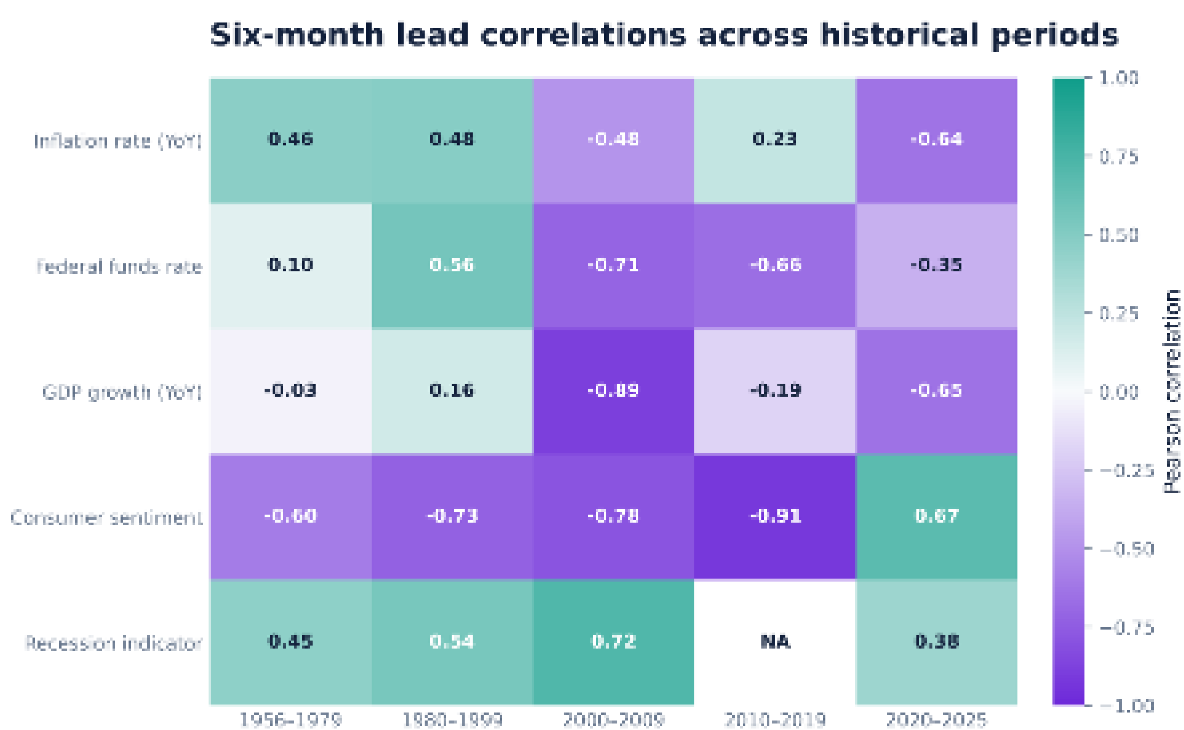
Bars report the mean absolute Pearson correlation across the same-month, three-month, six-month, and twelve-month comparisons.



The ranking uses absolute values because a negative relationship can be strong. Consumer sentiment ranks first, while recession status ranks second. The recession result is historically informative but not a clean real-time predictor because NBER recession dates are assigned retrospectively.

Figure 6. Six-month lead correlations across historical periods

Period-specific coefficients test whether the full-sample relationships remain stable across economic environments.



Consumer sentiment was strongly negative from 1956 through 2019, ranging from –0.60 to –0.91 at a six-month lead, but became positive (0.67) during 2020–2025. GDP growth, inflation, and interest-rate relationships also changed sign or magnitude across periods. The NA entry for recession status in 2010 –2019 occurs because there were no recession months in that interval, leaving the binary variable with no variation.

8. Discussion

8.1 Answer to Research Question 1

> **Answer**
> Yes, but only with a horizon-qualified conclusion. Three-month unemployment is forecast with strong predictive performance, six-month unemployment with moderate performance, and twelve-month unemployment with limited incremental value over simple baselines.

The three-month result is the clearest operational finding. A 0.349-point MAE means the typical monthly forecast error is roughly one-third of a percentage point, although RMSE rises to 1.020 because rare crisis errors are much larger. The six-month model retains more than half of test-period variance and improves RMSE over both baselines, making it potentially useful for medium-term planning. The twelve-month model explains only about one-fifth of test variation and does not outperform the unemployment-only baseline, so it should not be presented as a reliable one-year prediction.

8.2 Answer to Research Question 2

> **Answer**
> Consumer sentiment showed the strongest overall historical relationship with current and future unemployment. The relationship was negative and strengthened from r = –0.351 in the same month to r = –0.496 at twelve months. Recession status ranked second, but its retrospective construction limits real-time use. No external indicator was stable across every historical period.

This conclusion is distinct from the machine-learning results. The forecasting models used all 35 engineered predictors, whereas Research Question 2 ranks only the five external macroeconomic indicators using historical correlation. Consumer sentiment showed the strongest full-sample external relationship: weaker confidence historically aligned with higher unemployment several months later. Recession status also had a meaningful positive relationship, but official NBER dates are retrospective. GDP growth was modestly negative, while inflation and the federal funds rate became more positively associated with unemployment at longer leads.

Stability analysis qualifies this ranking. At a six-month lead, sentiment correlations were −0.60, −0.73, −0.78, and −0.91 across the first four historical periods, but +0.67 during 2020–2025. Other indicators also changed direction or magnitude. Consumer sentiment should therefore be described as the strongest overall relationship and the most consistent across forecast horizons in the full sample, not as a universal relationship across every economic regime. These interpretations agree with the reported full-sample and period-specific correlation results.

8.3 Why the selected algorithms differ

Ridge Regression was selected at three months because the regularized linear model achieved the lowest validation RMSE and reduced instability among correlated predictors. Extra Trees was selected at six and twelve months, indicating that nonlinear tree ensembles performed better on the corresponding validation periods. However, the weak twelve-month test performance shows that additional model flexibility did not produce a dependable one-year forecast.

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9. Limitations and Future Work

9.1 Limitations tied to the submitted analysis

- Retrospective recession information: NBER recession dates are assigned after turning points occur. RecessionIndicator and RecessionRiskScore therefore limit real-time deployment claims and may improve historical backtests using information not contemporaneously available.
- Composite-feature dependence: EconomicStressIndex and several ratios include unemployment or closely related transformations. Their feature importance is not independent evidence from outside the target series.

- Structural breaks: COVID-19 produced errors far outside normal historical behavior. Models trained on prior relationships cannot anticipate shocks that are absent from the predictor history.
- National aggregation: U.S. averages conceal differences by state, industry, occupation, age, race, and educational attainment. The model cannot support local or subgroup decisions.
- Revisions and data vintage: The dataset uses revised historical values, and quarterly GDP is carried across three months. A true real-time evaluation would use release-aligned historical vintages.
- Single split: One long test period provides an interpretable historical comparison but does not show how model rankings vary across repeated rolling-origin evaluations.
- No formal prediction intervals: Point forecasts do not communicate the widening uncertainty at six and twelve months.
- Correlation and lead-lag results are associational: coefficient magnitude and direction describe historical co-movement, not causal effects. Relationships can also change across regimes, as shown by the reversal of the six-month consumer-sentiment correlation during 2020–2025.
- Time-series dependence: monthly observations are serially correlated, so conventional correlation p-values can appear more precise than they are. The report emphasizes coefficient magnitude, direction, and period-to-period stability rather than statistical significance alone.

9.2 Future work

1. Create a strictly real-time feature set that removes retrospective recession labels and uses contemporaneously available proxies such as claims, payroll growth, vacancies, and real-time recession indicators.
2. Implement rolling-origin cross-validation and report the distribution of MAE and RMSE across historical windows rather than one validation period.
3. Build vintage-aware data from ALFRED or release archives to measure the effect of revisions.
4. Add prediction intervals through quantile regression, conformal prediction, or bootstrapped ensembles.
5. Compare national models with state- and industry-level forecasts where sufficiently long, consistently defined data are available.
6. Test dedicated time-series approaches, including autoregressive distributed lag models, dynamic factors, and recurrent or transformer-based sequence models, while retaining simple baselines.
7. Estimate rolling correlations and distributed-lag or autoregressive models to test whether the consumer-sentiment relationship remains after controlling for unemployment persistence, inflation, interest rates, GDP growth, and serial correlation.

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10. Conclusions and Recommendations

The results answer the two research questions separately. For **Research Question 1**, publicly available macroeconomic data supported useful short-horizon unemployment forecasting when evaluated chronologically. Ridge Regression was selected for the three-month horizon and achieved test MAE 0.349, RMSE 1.020, and R^2 0.720. The six-month Extra Trees model retained moderate predictive value, with RMSE 1.277 and R^2 0.563. At twelve months, performance declined substantially and the selected model provided almost no improvement over persistence and performed slightly worse than the unemployment-only baseline. The twelve-month forecast should therefore be treated as exploratory rather than a reliable one-year prediction.

For **Research Question 2**, consumer sentiment showed the strongest overall historical relationship with current and future unemployment. Its correlation strengthened from $r = -0.351$ in the same month to $r = -0.496$ at twelve months, indicating that weaker sentiment historically aligned with higher future unemployment. Recession status ranked second, although its retrospective construction limits its value as a real-time indicator. GDP growth showed a modest inverse relationship, while inflation and the federal funds rate became more positively associated with unemployment at longer leads. Because these relationships changed across historical periods—particularly after 2020—they should be interpreted as regime-dependent historical associations rather than causal or universally stable effects.

10.1 Recommendations for Use Cases

- Use the three-month Ridge Regression model as the primary forecasting result and the six-month Extra Trees model as a secondary planning tool.
- Present the twelve-month forecast as a broad scenario only, not as evidence of dependable one-year prediction.
- Use consumer sentiment as a contextual early-warning indicator, but not as a standalone forecasting or policy signal.
- Do not treat the NBER recession indicator as a real-time predictor because recession dates are assigned retrospectively.
- Retain persistence and unemployment-only baselines when reporting complex-model performance.
- Continue selecting models using validation performance rather than changing the selected model after inspecting test results.
- Reassess correlations and model performance across rolling historical windows because indicator relationships can change across economic regimes.

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11. Reproducibility

Table 12. Reproducibility checklist

Component	Implementation
Language	Python 3
Core packages	pandas, NumPy, scikit-learn, XGBoost, Matplotlib, SciPy, openpyxl, and Joblib
Input	capstone_capstone_plus_final.xlsx
Principal outputs	Model, baseline, prediction, feature-importance, lead-lag correlation, indicator-summary, period-stability, and recession-comparison CSV files; Joblib models; PNG/SVG figures
Repository	Deliverables/M5
Python Scripts	All Python scripts, including the separate RQ2 correlation analysis, are available in the Deliverables/M5 directory of the GitHub repository.

The numerical results presented in this report were taken directly from the CSV files generated during the submitted modeling and correlation-analysis runs. These files contain model comparisons, selected models, baseline comparisons, test predictions, feature importance, lead-lag correlations, indicator rankings, historical-period stability results, and recession-versus-expansion comparisons.

The full set of Python scripts required to reproduce both research questions is available in the Deliverables/M5 directory of the project GitHub repository. The RQ2 script reads the same processed Excel dataset and reproduces Pearson and Spearman correlations, lead-lag tables, period-stability analysis, and the figures used in Sections 7.6 and 8.2.

Results may change if the input data, feature definitions, date ranges, random seeds, hyperparameters, package versions, or model configurations are modified. Any such modification should be documented and treated as a new analysis version rather than a direct reproduction of the submitted results.

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